

A YOUNG KIM

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EDUCATION

KAIST, M.S. in Data Science | Advisor: Prof. Mun Yong Yi Sep 2024 – Aug 2026 (expected)

University of California, San Diego, B.S. in Probability and Statistics Sep 2016 – Dec 2019

PUBLICATIONS

Every Answer Has Its Own Path: Agentic Routing for Retrieval-Augmented Table Question Answering. *First author, EMNLP 2026.*

Distilling LLM Reasoning into Dense Encoders: Bridging the Accuracy-Efficiency Gap in Recommendation. *ACL 2026 Findings*. [\[paper\]](#)

SAFE: A Plug-and-Play Framework for Prompt Optimization in Text-to-Image Safety. *EMNLP 2026.*

DRCoD: Toward Robust Continual Learning of Diffusion Models for Tire Manufacturing Prototyping. *CVPR 2026 Workshop*. [\[paper\]](#)

RESEARCH EXPERIENCE

Graduate Researcher, KAIST: Retrieval-Augmented Generation and Agentic LLM Systems Sep 2024 – Aug 2026

- Built **ATR**, a first-author agentic framework for QA over hybrid table-text documents that routes each sub-query to one of four execution primitives and re-routes when a verifier rejects the evidence. A single configuration **leads on token F1 across three benchmarks** without per-dataset tuning.
- Designed a five-view offline index (text chunks, table chunks, schema, cell values, relational DB) and a value-linking step that grounds every entity mention against the cell index before SQL execution, ruling out column and value hallucination by construction.
- Distilled the routing policy into a lightweight DistilBERT student that reproduces **98.6%** of teacher routing decisions at zero LLM cost per sub-query, transferring zero-shot to two unseen benchmarks.
- Co-developed **R2END** (ACL 2026 Findings): distilled LLM reasoning into a compact dense encoder for recommendation, a **29% average accuracy gain** over the best baselines across four datasets at an order of magnitude lower inference latency.
- Stack**: benchmarked closed- and open-weight backbones (Gemini, Claude, Llama, Qwen) served via Vertex AI and self-hosted vLLM, where a self-hosted Llama matched closed-weight accuracy at zero API cost; BGE-M3 retrieval with cross-encoder reranking; MySQL execution backend; PyTorch, Hugging Face.

Graduate Researcher, KAIST x Hankook Tire: Multimodal Diffusion for Industrial Generation Sep 2024 – Dec 2025

- Built multimodal diffusion pipelines fusing tabular specs, 1D sequences, and 2D images for industrial prototyping (**TAMIG**), plus a reference-free evaluation framework (**RFEM**) automating quality assessment without ground truth.

Research Intern, Computer Vision Lab, Korea University Feb 2024 – Jul 2024

- Researched novel view synthesis and 3D scene representation (**NeRF**, **3D Gaussian Splatting**) and AI watermarking of reconstructed scenes.

INDUSTRY EXPERIENCE

Founder & AI Engineer, Rimbic, Inc. Jun 2024 – Jun 2025

- Built an end-to-end AI smart speaker: real-time voice agent (OpenAI Realtime API, Agora SDK) with fine-tuned **Stable Video Diffusion** on edge hardware (Jetson Nano, Raspberry Pi).
- Secured **47M KRW (~\$35K) non-dilutive government funding** through competitive selection in the Pre-Entrepreneurship Package program; led product, engineering, and GTM end to end.

Digital Consultant, KPMG Korea Dec 2021 – Nov 2023

- Designed and deployed a customer segmentation and product recommender system leveraging MyData for a top-tier Korean bank, scaled to **3.55M+ customers** in production.
- Built **time-series forecasting models** in TensorFlow for six agricultural commodities on NongHyup's N-Hub platform, querying Apache Hadoop through Hive and Impala.
- Designed data marts for bank fund-management reporting; ran store clustering and market-basket analysis across the HanaroMart retail network.

Software Engineer, LG Electronics North America Mar 2020 – Nov 2020

- Automated **100+** regression test cases in Python across network, connectivity, and battery scenarios for LG mobile devices.

TECHNICAL SKILLS

LLM / NLP: retrieval-augmented generation, agentic systems (sub-query routing, tool use, verifier-based escalation), knowledge distillation, LLM fine-tuning, table QA, text-to-SQL, recommendation

Frameworks & Serving: PyTorch, Hugging Face (Transformers, Datasets), vLLM, TensorFlow | **Retrieval & Data**: BGE-M3, MySQL, Apache Hadoop (Hive, Impala)

Generative & Vision: diffusion models (Stable Diffusion, SVD), multimodal fusion, NeRF, 3D Gaussian Splatting | **Programming**: Python, SQL, Java, R, Shell